Answer Key & Teacher Notes

Session 1 — Rational Numbers on a Number Line · Session 2 — Opposites of Rational Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Rational Numbers on a Number Line

TALK PROMPTS

- How do you decide how many equal parts to mark between two integers?
- Why is −3.5 to the left of −3, even though 3.5 is greater than 3?

PARTNER TASK

Lab Thermometer. One partner plots −2.5 and 2.5 on a −4 to 4 number line marked in halves; the other proves they are opposites using distance from 0. Switch roles with −1.25 and 1.25.

WATCH FOR

- Counting tenths to the right of −3 to plot −3.4 (the point belongs between −4 and −3).
- Splitting the gap by the numerator (7/4: split into 4 parts, not 7).
- Thinking a whole number such as 7 is not rational.

SESSION 2 · TEACHER NOTES

Opposites of Rational Numbers

TALK PROMPTS

- Why does taking the opposite twice bring a number back to where it started?
- What does the sign in −(−4.5) mean in words?

PARTNER TASK

Card Flips. One partner writes a fraction or decimal on a card; the other names its opposite and its distance from 0. Flip again and predict the result before checking on a number line.

WATCH FOR

- Believing the opposite of a number is always negative.
- Reading −(−a) as "more negative" instead of "the opposite of −a."
- Thinking 0 has no opposite (it is its own opposite).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.6c	1	2	1	1	2	2	2	2	2	1	1
S1 · Form B	6.NS.C.6c	1	2	1	1	2	2	2	2	2	1	1
S2 · Form A	6.NS.C.6a	1	2	2	1	2	2	1	2	2	1	1
S2 · Form B	6.NS.C.6a	1	2	2	1	2	2	1	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

rational number: A number that can be written as a **fraction** of two integers.

integer: A whole number or its **opposite**.

number line: A line where numbers **increase** from left to right.

- To plot 7/4, split each whole into **4** equal parts and count **7** parts from 0.
- Negative numbers sit to the **left** of 0 on a horizontal line and **below** 0 on a vertical line.

Example: −2.75 is between the integers **−3** and **−2**. Split the gap between −3 and −2 into **4** equal parts (fourths). Start at −2 and count **3** fourths to the left. Answer: −2.75 is one fourth to the **right** of −3.

YOU TRY

1. −1.5 halfway between −2 and −1; 2.25 one fourth to the right of 2

2. a. −4 and −3 b. 2 and 3

EXIT TICKET

1. a. −5 and −4 b. 2 and 3

2. −0.75 three fourths left of 0; 1.5 halfway between 1 and 2

3. **B** (−2.25)

PRACTICE FORM A

1. a. −5 and −4 b. 0 and 1 c. −3 and −2 d. −3 and −2

2. a. 3.5, −0.75, 0.5, −0.2 b. 3.5 halfway between 3 and 4; −0.75 three fourths left of 0; 0.5 halfway between 0 and 1; −0.2 two tenths left of 0

3. 2.5 degrees (both readings are 2.5 from 0)

4. −1.25

5. 7/1; Lena is not right, so 7 is rational

6. −0.45 (−11/4 = −2.75 is between −3 and −2; 1.3 is between 1 and 2)

7. 1.6 − 0.4 = 1.2 m

8. a. −12.5 m b. −13 and −12

9. Counting right of −3 moves toward 0, which gives −2.6. −3.4 is 0.4 to the LEFT of −3, so it is between −4 and −3.

10. **B** (−3 and −2)

11. **A** (−4.5)

Self-check word **NUMBER**

PRACTICE FORM B

1. a. −4 and −3 b. 0 and 1 c. −3 and −2 d. −4 and −3

2. a. 2.5, −0.5, 1.25, −0.3 b. 2.5 halfway between 2 and 3; −0.5 halfway between −1 and 0; 1.25 one fourth past 1; −0.3 three tenths left of 0

3. 3.5 degrees (both readings are 3.5 from 0)

4. −1.75

5. −4/1; Jalen is not right, so −4 is rational

6. −0.35 (−9/4 = −2.25 is between −3 and −2; 1.6 is between 1 and 2)

7. 2.1 − 0.6 = 1.5 m

8. a. −8.5 m b. −9 and −8

9. Counting right of −3 moves toward 0, which gives −2.4. −3.6 is 0.6 to the LEFT of −3, so it is between −4 and −3.

10. **C** (−2 and −1)

11. **C** (0.25)

Self-check word **POINTS**

GUIDED NOTES

opposites: Two numbers the same **distance** from 0, on different sides of 0.

opposite of the opposite: Taking the opposite **twice** gives back the number you started with.

zero: The only number that is its **own** opposite.

- The symbol −(4.5) means the **opposite** of 4.5, which is −4.5.
- The opposite of a negative number is **positive**.

Example: −5.3 is **5.3** units to the left of 0. Its opposite is the same distance on the **right** side of 0. So −(−5.3) = **5.3**. Answer: −(−5.3) = **5.3**

YOU TRY

1. a. −2/9 b. 0.6 c. 0

2. B = 1.25

EXIT TICKET

1. a. 3.2 b. 7/9 c. −1.4

2. −6.5

3. No. The opposite of −4 is 4, which is positive.

PRACTICE FORM A

1. a. 2.5 b. −7/10 c. 1/4 d. 0 e. −4.5 f. 4.5

2. a. 4.5, 7.25, 11.5, 0.8 b. 7.25 m

3. −7/8 (first flip 7/8, second flip −7/8)

4. 18.6 dollars

5. 4.8

6. a. −6.75 ft b. 6.75 ft each

7. −6.35

8. a. 4.5, −4.5, 0.5 b. −0.5

9. The first flip gives −8.6; the second flip goes back across 0 to 8.6. Two opposites return to the start, so the answer is 8.6.

10. **B** (3.8)

11. **C** (5/6)

Self-check word **REFLECT**

PRACTICE FORM B

1. a. 3.5 b. −3/10 c. 3/4 d. 0 e. −6.2 f. 6.2

2. a. 3.5, 6.25, 10.5, 1.2 b. 6.25 m

3. −5/8 (first flip 5/8, second flip −5/8)

4. 24.3 dollars

5. 3.6

6. a. −5.25 ft b. 5.25 ft each

7. −8.15

8. a. 3.5, −3.5, 0.25 b. −0.25

9. −(−7.4) means the opposite of −7.4. The opposite of a negative number is positive, so −(−7.4) = 7.4.

10. **B** (2.6)

11. **C** (3/7)

Self-check word **FLIPPED**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · RATIONAL NUMBERS ON A NUMBER LINE

1. A = −1.75 (−1 3/4) B = −0.5 (−1/2) C = 1.25 (1 1/4)

2. −2.5 halfway between −3 and −2; 1/2 halfway between 0 and 1; −0.75 three fourths left of 0; 1 1/4 one fourth right of 1.

3. a. −5 and −4 b. 3 and 4 c. −1 and 0 d. −3 and −2

4. a. −3.50 b. 2 1/2 c. −4.5

5. a. frog 1.5, fish −2.25 b. the frog (1.5 units from 0 is less than 2.25 units from 0)

6. −1.5 is 1.5 units left of 0, so it is past −1. It belongs halfway between −2 and −1.

SHEET 2 · OPPOSITES OF RATIONAL NUMBERS

1. a. −4.8 b. 2/3 c. 0.05 d. 0 e. 1.6 f. 5/8

2. Q at 2.25 (2 1/4), one fourth to the right of 2.

3. 12.5 points; it is 12.5 points from 0 (same distance as −12.5)

4. a. M = −3.75 (−3 3/4) b. N = 3.75 (3 3/4)

5. −(−2/5) means the opposite of −2/5. The opposite of a negative number is positive, so −(−2/5) = 2/5.

6. **B** (7/8)

7. **C** (3.4)

October 2026 conceptual additions — teacher key

Session 1 · Rational Numbers on a Number Line

Each interval is 0.25. −1.75 lies one interval right of −2, or three intervals left of −1.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Each interval is 0.25. −1.75 lies one interval right of −2, or three intervals left of −1.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Opposites of Rational Numbers

−1.5 and 1.5, equal distances on opposite sides of zero. −(−1.5)=1.5; taking the opposite again returns −1.5.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: −1.5 and 1.5, equal distances on opposite sides of zero. −(−1.5)=1.5; taking the opposite again returns −1.5.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.